

# 11. Matching Estimators

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# It's a Match!

You and MHE have liked each other!



Pixiz

Credit: Created using Pixiz (<https://en.pixiz.com/template/It-s-a-Match-Tinder-mockery-with-customizable-text-3150>)

# Where are we? Where are we going?

- The recurring question of this course: **where do we find good controls?**
  - Units *randomized* to receive control (RCT)
  - Units with *similar* values of covariates (SOO / reg. adj.)
  - Units with *opposite* value of some instrument (IV)
  - Units *near a discontinuity* in treatment assignment (RDD, W13)
  - Exploit two possible sources of variation for identification:
    - Exploit **cross-sectional** variation in treatment.
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    - Exploit variation in treatment **within a unit over time** (before/after)
- Can we make our identification strategies work *better*?
  - $\rightsquigarrow$  **matching** or **weighting**
  - Today: refine the SOO branch by pruning or re-weighting observational controls.

# 1/ Matching Estimators

# The Problem with Regression

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- Recall the **imputation** estimators with regression (ECI W5).

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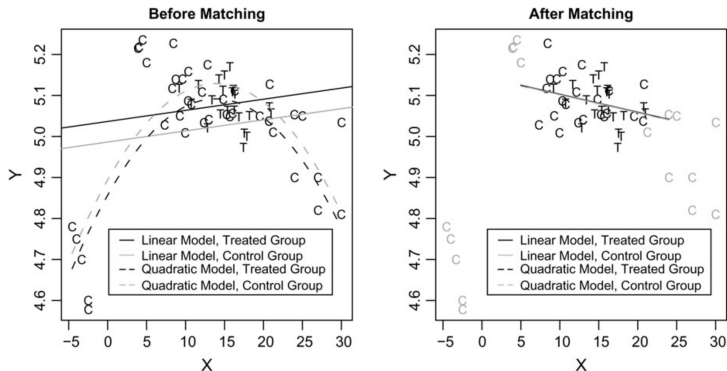
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- Common solution: use a parametric model for  $\hat{\mu}_0(\mathbf{x}_i)$ .
  - For example, could assume it is linear:  $\mu_0(\mathbf{x}) = \mathbf{x}'\beta$
  - Regression, MLE, Bayes, etc.
  - But this model might be wrong  $\rightsquigarrow$  wrong causal estimates.

# Model Dependence



Source: *Figure 1* in Ho, Daniel E., Kosuke Imai, Gary King, and Elizabeth A. Stuart. "Matching as nonparametric preprocessing for reducing model dependence in parametric causal inference." *Political analysis* 15, no. 3 (2007): 199-236.

# What is Matching?

- **Matching** is a nonparametric imputation estimator:

$$\hat{\tau}_m = \frac{1}{n_1} \sum_{i=1}^n D_i \left( Y_i - \frac{1}{M} \sum_{j \in \mathcal{J}_M(i)} Y_j \right)$$

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- Matching has strong advantages:
  1. Reduces dependence of estimates on parametric models.
  2. Reduces model-based extrapolation.
  3. Makes counterfactual comparisons more transparent.
- What matching isn't: a solution for selection on unobservables.
  - Matching is an **estimation** technique, not an identification strategy.

# Types of Matching

- Assumptions:
  - No unmeasured confounders:  $D_i \perp\!\!\!\perp (Y_i(0), Y_i(1)) \mid \mathbf{x}_i$
  - Overlap/positivity:  $0 < \mathbb{P}(D_i = 1 \mid \mathbf{x}_i = x) < 1$

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- Exact matching:** choose matches that have the same value of  $\mathbf{X}_i$ .
  - $\mathcal{J}_M(i)$  is a random set of  $M$  control units with  $\mathbf{X}_j = \mathbf{X}_i$
  - Covariate distribution in treated and matched controls exactly the same:

$$\begin{aligned}\widehat{\mathbb{P}}(\mathbf{X}_i = \mathbf{x} \mid D_i = 1) &= \widehat{\mathbb{P}}(\mathbf{X}_j = \mathbf{x} \mid D_j = 0, j \text{ is matched}) \\ \rightsquigarrow \mathbb{E}[Y_i(0) \mid D_i = 1] &= \mathbb{E}[Y_j(0) \mid D_j = 0, j \text{ is matched}]\end{aligned}$$

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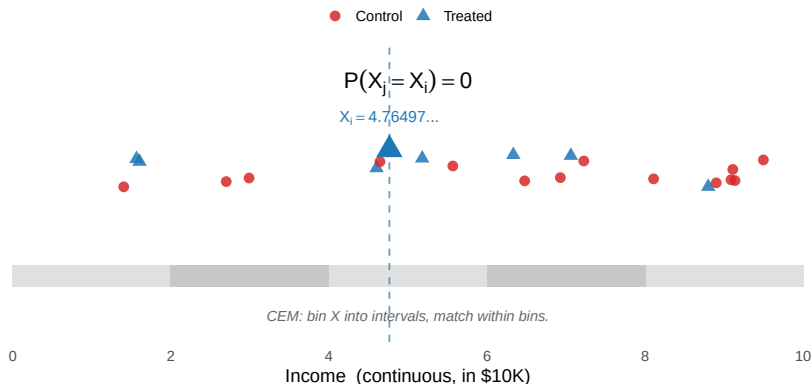
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- **Coarsened exact matching** (Iacus et al, 2011)
  - Discretize and group covariates into substantively meaningful bins
  - Exact match on these bins  $\rightsquigarrow$  accounts for interactions
  - Have to drop treated units in bins with no controls  $\rightsquigarrow$  changes estimand
  - Allows you to control bias/variance tradeoff through coarsening

# Continuous X: Exact Matching Fails

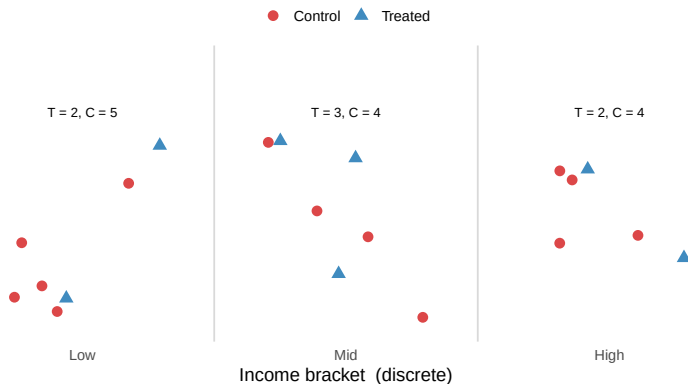
- Say  $\mathbf{X}$  = income (continuous). Then  $X_j \neq X_i$  “almost surely” (a.s.), i.e.,  $\mathbb{P}(X_j = X_i) = 0$ :



- Fix: **coarsen**  $\mathbf{X}$  into bins (CEM) and match within each bin.

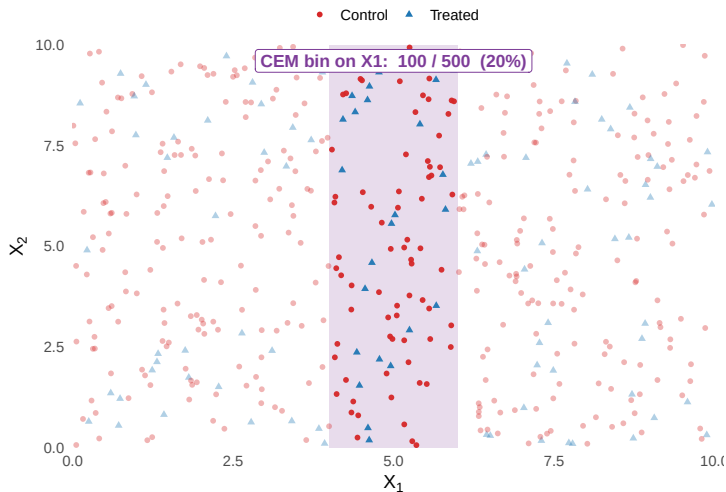
# CEM: Bin Continuous X into Brackets

- Income brackets {Low, Mid, High} are a CEM coarsening. Pair each treated with a control in the **same bracket**:



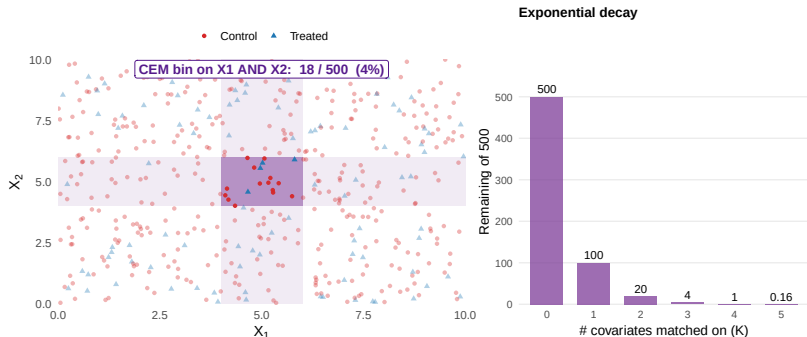
# Curse of Dimensionality: CEM on One Covariate

- Coarsen  $X_1$  into bins; keep observations in one bin:



# Curse of Dimensionality: CEM on Multiple Covariates

- Now require closeness on *both*  $X_1$  and  $X_2$  (intersection of two bins):



- Matching breaks when a treated unit's bin has no controls. As  $K$  grows, this becomes increasingly common.

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- **Mahalanobis distance:**

$$D(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{(\mathbf{x}_i - \mathbf{x}_j)' \hat{\Sigma}^{-1} (\mathbf{x}_i - \mathbf{x}_j)}$$

- $\hat{\Sigma}$  is the estimated variance-covariance matrix of the observations:

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

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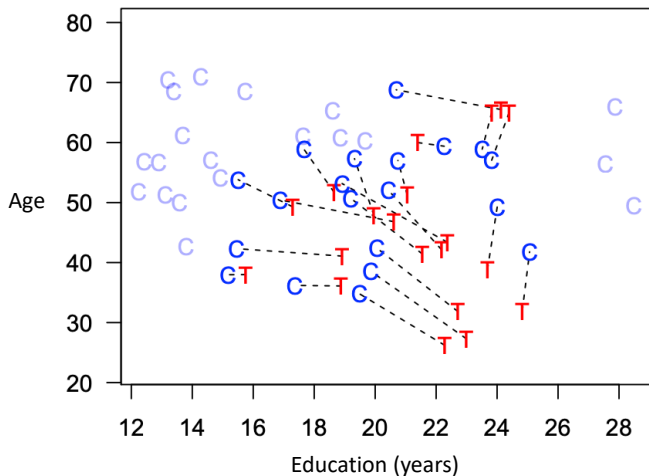
- **Estimated propensity score** (Rosenbaum and Rubin, 1983):

$$D(\mathbf{x}_i, \mathbf{x}_j) = |\widehat{\pi}(\mathbf{x}_i) - \widehat{\pi}(\mathbf{x}_j)| = |\widehat{\mathbb{P}}(D_i = 1 \mid \mathbf{x}_i) - \widehat{\mathbb{P}}(D_j = 1 \mid \mathbf{x}_j)|$$

- Some use linear predictor:  $\text{Dist}_{ij} = |\text{logit}(\widehat{\pi}(\mathbf{x}_i)) - \text{logit}(\widehat{\pi}(\mathbf{x}_j))|$

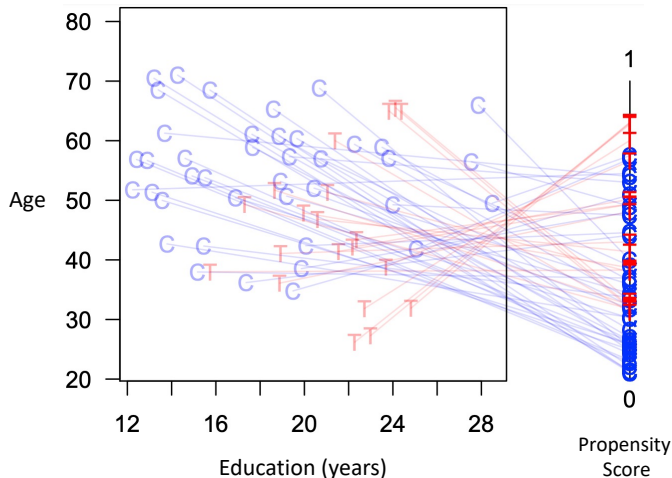
# Mahalanobis Distance Matching

- **Mahalanobis Distance Matching:**
  - Prune observations where  $\text{Dist}_{ij} > \text{caliper}$  ( $\leadsto$  caliper? = cutoff point)



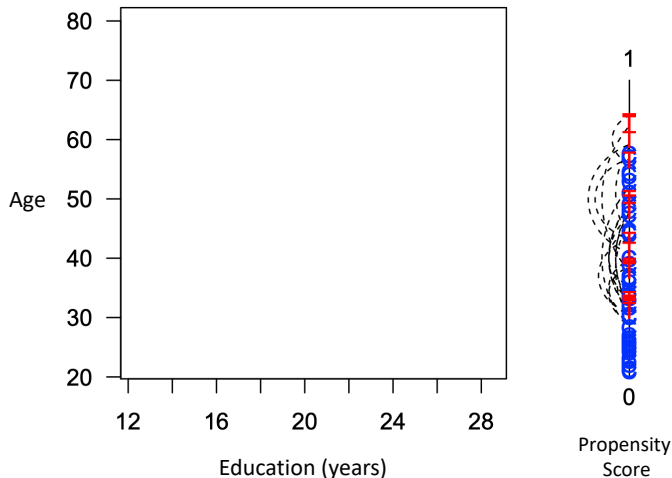
Credit: Figure from Gary King's talk on "The Balance-Sample Size Frontier in Matching Methods for Causal Inference," at University of Michigan, January 24, 2014.

# Propensity Score Matching Illustrated



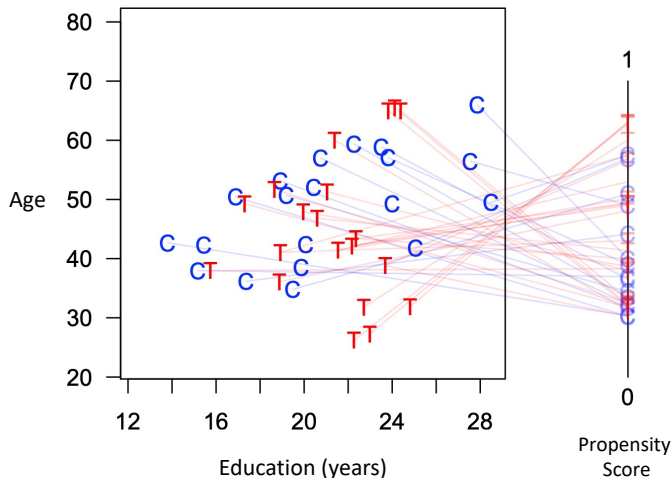
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- **With or without replacement:** can the same control be matched to multiple treated?
  - **With** replacement: each treated picks its closest control independently  
↪ better matches, order doesn't matter. But the same control reappears in many pairs ↪ pairs are not independent ↪ SE must **cluster on the matched control** (Abadie & Imbens, 2006).
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- **Caliper:** drop poor matches?
  - Only keep matches below a distance threshold,  $D(\mathbf{x}_i, \mathbf{x}_j) \leq c$ 
    - Rosenbaum and Rubin (1985): use  $c$  size equiv. to  $0.25 \times \text{sd of the PS}$ .
  - Reduces imbalance, but if you drop treated units, estimand changes.
    - ↪ If we drop treated units, what are we estimating other than the ATT?

# More on Propensity Scores

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- $\rightsquigarrow$  “propensity score tautology”.

# Assessing Balance

- Goal of matching is to maximize balance:  $\hat{F}_1(\mathbf{x}) \approx \hat{F}_{0,\mathcal{J}}(\mathbf{x})$ 
  - Joint distribution of  $\mathbf{X}_i$  is similar between treated and matched controls.
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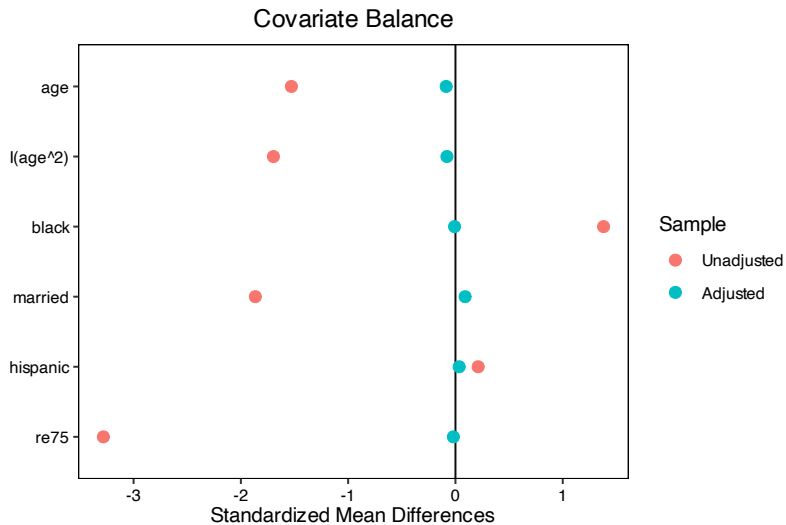
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- Hypothesis tests for balance are problematic:
  - Dropping units can lower power ( $\uparrow$  p-values) without a change in balance.

# Balance Diagnostics: Love Plot



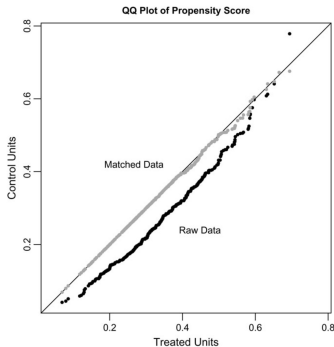
# Balance Diagnostics: Table and Density

|                                      | Nonadopters |         | Adopters |         | Differences in Means |
|--------------------------------------|-------------|---------|----------|---------|----------------------|
|                                      | Mean        | Std dev | Mean     | Std dev |                      |
| Customer spending:                   |             |         |          |         |                      |
| Purchase amount (monetary)           | 21.133      | 14.251  | 21.207   | 15.407  | -0.074               |
| Number of transactions (frequency)   | 1.318       | 0.578   | 1.321    | 0.536   | -0.003               |
| Number of transactions per trip      | 1.161       | 0.450   | 1.149    | 0.422   | 0.012                |
| Days since last purchase (recency)   | 118.011     | 83.374  | 117.085  | 85.634  | 0.926                |
| Number of books purchased (quantity) | 1.819       | 1.173   | 1.848    | 1.256   | -0.029               |
| Maximum price of the books purchased | 12.915      | 5.782   | 12.957   | 6.311   | -0.042               |
| Consumption variety:                 |             |         |          |         |                      |
| Number of unique books               | 1.733       | 1.101   | 1.766    | 1.149   | -0.033               |
| Number of unique genres              | 1.272       | 0.511   | 1.286    | 0.512   | -0.014               |
| Number of unique authors             | 1.975       | 1.448   | 2.019    | 1.473   | -0.044               |
| Number of unique publishers          | 1.685       | 0.988   | 1.696    | 0.970   | -0.011               |
| Concentration on personal favorites: |             |         |          |         |                      |
| Genre concentration (norm. HHI)      | 0.907       | 0.161   | 0.903    | 0.156   | 0.004                |
| Author concentration (norm. HHI)     | 0.753       | 0.266   | 0.750    | 0.256   | 0.003                |
| Publisher concentration (norm. HHI)  | 0.788       | 0.239   | 0.791    | 0.228   | -0.003               |
| Best-seller purchases:               |             |         |          |         |                      |
| Shares of top 10 best sellers        | 0.076       | 0.219   | 0.074    | 0.201   | 0.002                |
| Shares of top 50 best sellers        | 0.147       | 0.298   | 0.142    | 0.272   | 0.005                |
| Usage of Sales Channels:             |             |         |          |         |                      |
| Share of in-store mobile sales       | 0.147       | 0.328   | 0.190    | 0.362   | -0.043               |
| Share of m-commerce sales            | 0.071       | 0.237   | 0.096    | 0.268   | -0.025               |
| Share of e-commerce sales            | 0.086       | 0.261   | 0.094    | 0.267   | -0.008               |
| Share of cash register sales         | 0.103       | 0.279   | 0.107    | 0.278   | -0.004               |
| Customer demographics:               |             |         |          |         |                      |
| Age                                  | 31.193      | 7.310   | 31.151   | 7.394   | 0.042                |
| Gender (male = 1)                    | 0.333       | 0.471   | 0.341    | 0.474   | -0.008               |

Note. Means and standard deviations of the observed variables, calculated in the 11-month pre-introduction period before February 1, 2015. The unit of analysis is an individual customer.

\* Significant at 10%; \*\* significant at 5%; \*\*\* significant at 1%.

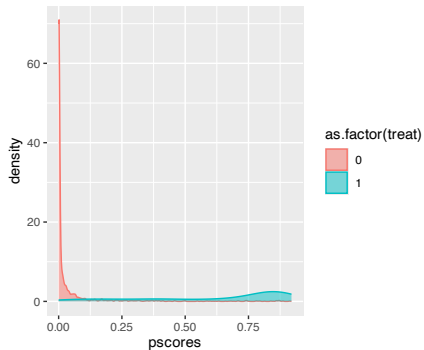
Source: Table 3 of Yoo et al. "Mobile Payment and In-Store Mobile Purchase Behavior"  
KAIST Working Paper Series



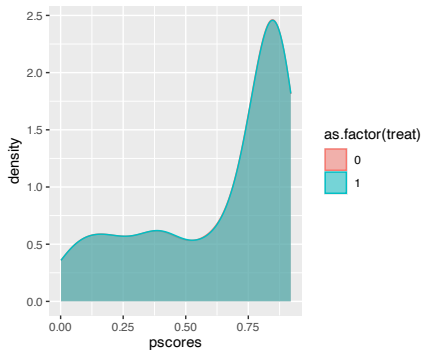
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# Balance Diagnostics: PS Overlap

Propensity Score Distribution Before Matching



Propensity Score Distribution After Matching



# Example: LaLonde Dataset

- The effectiveness of a job training program (National Supported Work Demonstration; NSW) on wage increases.
  - The federal government instituted a randomized evaluation of this program.
  - How well the result may be recovered when the experimental controls are replaced with a set of observational controls (Population Survey of Income Dynamics; PSID)?
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- **Problem:** Imbalances between the experimental and observational data  $\rightsquigarrow$  use matching.

# Example: LaLonde dataset

- **Data:**

- Treated: 297 units from NSW
- Control: 2490 units from PSID
- Treatment: Participation in the job training program (**treat**)
- Outcome: 1978 earnings (in dollars; **re78**)
- Pre-treatment covariates: age, race, marriage, past earnings, past employment

# Example R Codes: Data Import

- Import and process data:

```
1 > pacman::p_load(tidyverse, broom, cobalt, Matching, MatchIt)
2
3 > lalonde_nsw <- haven::read_dta(url("http://www.nber.org/~rdehejia/data/nsw.dta"))
4 > PSID_obs <- haven::read_dta(url("http://www.nber.org/~rdehejia/data/psid_controls.dta"))
5
6 > lalonde_ECI <- full_join(lalonde_nsw |>
7   filter(treat == 1),
8   PSID_obs); lalonde_ECI
9
10 # A tibble: 2,787 x 11
11   data_id   treat   age education black hispanic married nodegree re75 re78
12   <chr>     <dbl> <dbl>     <dbl> <dbl>   <dbl>   <dbl>   <dbl> <dbl> <dbl>
13 1 Lalonde S...   1    37         11     1       0       1       1     0 9930.
14 2 Lalonde S...   1    22          9     0       1       0       1     0 3596.
15 3 Lalonde S...   1    30         12     1       0       0       0     0 24909.
16 # ... 2,784 more rows
```

# Example R Codes: Balance Before Matching

- Assessing balance before matching:

```
1 > bal.tab(x = lalonde_ECI |> dplyr::select(age:re78),
2         treat = lalonde_ECI$treat, continuous = "std", binary = "std")
3
4 Note: `s.d.denom` not specified; assuming "pooled".
5 Balance Measures
6           Type Diff.Un
7 age      Contin. -1.1662
8 education Contin. -0.6862
9 black     Binary  1.3222
10 hispanic  Binary  0.2554
11 married   Binary -1.9513
12 nodegree  Binary  0.9409
13 re75      Contin. -1.5662
14 re78      Contin. -1.2939
15
16 Sample sizes
17       Control Treated
18 All      2490      297
```

# Example R Codes: Propensity Score Matching

- Estimate propensity score (logit) and match 1-to-1 NN:

```
1 # Propensity score
2 > pcores <- glm(treat ~ age + I(age^2) + black + married + hispanic + re75,
3               family = binomial(), data = lalonde_ECI)$fitted.values
4
5 # Conduct one-to-one nearest neighbor propensity score matching
6 > require(Matching)
7 > match_ps <- Match(Y=lalonde_ECI$re78, Tr=lalonde_ECI$treat,
8                   X=pcores, M=1, replace = TRUE)
9 > c(est = as.numeric(match_ps$est), se = match_ps$se)
10      est      se
11 -2030.895 1300.975
```

# LaLonde: ATT Estimates Compared

- Three views of the same NSW-vs-PSID effect on 1978 earnings:

| Method                                                        | ATT (\$) | SE (\$) |
|---------------------------------------------------------------|----------|---------|
| Experimental benchmark (NSW only, $n_T = 297$ , $n_C = 425$ ) | +886     | 488     |
| Naive observational (PSID controls, no matching)              | -15,578  | 509     |
| PSM, 1-to-1 NN with replacement                               | -2,031   | 1,301   |

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- Naive obs flips the *sign*: -\$16K vs. +\$886 in the RCT.
- PSM closes  $\approx 87\%$  of the bias gap, but not all of it.
- Residual gap: selection on *unmeasured* confounders (motivation, networks) + finite-sample matching is imperfect on observables.



## On to the Presentations & Discussions!

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# Appendix

# Bias of Inexact Matching

- To show the bias on matching, focus on finding a single control match.
- Let  $j(i)$  be the matched control for unit  $i$ , the bias is:

$$\mathbb{E}[Y_j \mid D_i = 1, \mathbf{x}_i, \mathbf{x}_j] - \mathbb{E}[Y_i(0) \mid D_i = 1, \mathbf{x}_i] = \underbrace{(\mu_0(\mathbf{x}_i) - \mu_0(\mathbf{x}_{j(i)}))}_{\text{unit-level bias}}$$

- Bias is 0 if matching is exact since  $\mathbf{x}_i = \mathbf{x}_{j(i)}$
- Bias grows with **matching discrepancy**/imbalance.
- **Bias correction:** estimate  $\hat{\mu}_0(\mathbf{x})$  with regression and estimate bias.

$$\hat{Y}_i(0) = Y_{j(i)} - (\hat{\mu}_0(\mathbf{x}_i) - \hat{\mu}_0(\mathbf{x}_{j(i)}))$$

- Imputation of missing potential outcome now matching + regression.
- Generalizes easily to any number of matches.

# Sampling Variance of Matching Estimators

- Matching with replacement: cluster on the match.
  - Can either use clustered SEs or cluster bootstrap.
  - Valid for post-matching regression (Abadie and Spiess, 2021).
- Matching without replacement: more complicated.
  - Same control unit matched to multiple treated: no easy clustering.
  - $K_M(i)$  is the number of times a unit is used as a match.
- Assuming units are well-matched so bias can be ignored,

$$\mathbb{V}(\widehat{\tau}_m) = \frac{1}{n_1} \left( \underbrace{\mathbb{E} [(\tau(\mathbf{X}_i) - \tau_{\text{ATT}})^2 \mid D_i = 1]}_{\text{variance of CATE on treated}} + \underbrace{\mathbb{V}[\widehat{\tau}_m \mid \mathbb{X}, \mathbf{D}]}_{\text{conditional variance}} \right)$$

- Abadie and Imbens (2006) provide matching-based variance estimators.